

# Zhuoran Yu

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## Education

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### Ph.D. in Computer Science

Sep 2021 – May 2026

University of Wisconsin–Madison, Madison, WI

*Advisor: Yong Jae Lee*

### M.S. in Computer Science

Aug 2018 – Aug 2020

Georgia Institute of Technology, Atlanta, GA

*GPA: 4.0/4.0*

### B.Math in Computer Science and Applied Mathematics

Jan 2014 – Jun 2018

University of Waterloo, Waterloo, Canada

Graduated with Distinction & Dean's Honor List

Arthur Beaumont Memorial Scholarship

## Research and Professional Experience

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### Senior AI Research Scientist

Jun 2026 – Present

Autodesk, Inc., San Francisco, CA

- Conduct research and development at Autodesk Research AI Lab, focusing on foundation models for 3D content generation and multimodal design intelligence.
- Develop and evaluate generative AI methods for three-dimensional design and shape generation from multimodal inputs.

### Research Scientist Intern

May 2025 – Aug 2025

IBM Watson AI Lab, Cambridge, MA

- Contributed to the development and evaluation of Granite vision-language models, focusing on document and multimodal understanding.

### Research Scientist Intern

May 2024 – Jan 2025

Meta FAIR, Language and Communication Team, Seattle, WA / Remote

- Conducted research on data mixture strategies for early-fusion multimodal large language models.

### Research Scientist Intern

May 2023 – Nov 2023

Meta, CORE AI Team, Menlo Park, CA / Remote

- Developed synthetic-data-based approaches for improving visual recognition models using pre-trained diffusion models.

### Applied Scientist Intern

May 2022 – Aug 2022

AWS AI Rekognition, Seattle, WA

- Developed semi-supervised learning methods for 2D human pose estimation using denoised pseudo-heatmaps and uncertainty-guided pseudo-label selection.

## Publications

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\* denotes equal contribution

### DocHop: Benchmarking Out-of-domain Multi-hop Reasoning in Information-Dense Documents

Zhuoran Yu, Le Thien Phuc Nguyen, Jaden Park, Xinyi Gu, Zexue He, Soochahn Lee, Rogerio Feris,

Yong Jae Lee

*International Conference on Machine Learning (ICML)*, Seoul, South Korea, 2026.

**Large Language Model Teaches Visual Students: Cross-Modality Transfer of Fine-Grained Conceptual Knowledge**

Thomas Shih-Chao Liang\*, **Zhuoran Yu\***, Yong Jae Lee

*International Conference on Machine Learning (ICML)*, Seoul, South Korea, 2026.

**Revisiting Active Speaker Detection: An In-the-Wild Benchmark for Generalization and Robustness**

**Zhuoran Yu\***, Khoa Quang Nhat Cao, Yuwei Guo, Tu Ho Manh Pham, Tuan Tai Nguyen, Toan Ngo Duc Vo, Lucas Poon, Tuan Khai Nguyen, Soochahn Lee, Yong Jae Lee

*INTERSPEECH*, Sydney, Australia, 2026.

**See, Hear, and Understand: Benchmarking Audiovisual Human Speech Understanding in Multimodal Large Language Models**

Le Thien Phuc Nguyen\*, **Zhuoran Yu\***, Samuel Low Yu Hang, Subin An, Jeongik Lee, Yohan Ban, SeungEun Chung, Thanh-Huy Nguyen, JuWan Maeng, Soochahn Lee, Yong Jae Lee

*Findings of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR Findings)*, Denver, Colorado, 2026.

**LASER: Lip Landmark Assisted Speaker Detection for Robustness**

Le Thien Phuc Nguyen\*, **Zhuoran Yu\***, Yong Jae Lee

*IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Tucson, Arizona, 2026. Oral.

**Composition-Grounded Instruction Synthesis for Visual Reasoning**

Xinyi Gu, Jiayuan Mao, Zhang-Wei Hong, **Zhuoran Yu**, Pengyuan Li, Dhiraj Joshi, Rogerio Feris, Zexue He

*International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, 2026.

**DAVE: A VLM Vision Encoder for Document Understanding and Web Agents**

Brandon Huang, Hang Hua, **Zhuoran Yu**, Trevor Darrell, Rogerio Feris, Roei Herzig

*International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, 2026.

**How Multimodal LLMs Solve Image Tasks: A Lens on Visual Grounding, Task Reasoning, and Answer Decoding**

**Zhuoran Yu**, Yong Jae Lee

*Conference on Language Modeling (COLM)*, Montreal, Canada, 2025.

**CuRe: Cultural Gaps in the Long Tail of Text-to-Image Systems**

Aniket Rege, Zinnia Nie, Mahesh Ramesh, Unmesh Raskar, **Zhuoran Yu**, Aditya Kusupati, Yong Jae Lee, Ramya Korlakai Vinayak

*IEEE/CVF International Conference on Computer Vision (ICCV)*, Honolulu, Hawaii, 2025.

**Diversify, Don't Fine-Tune: Scaling Up Visual Recognition Training with Synthetic Images**

**Zhuoran Yu**, Chenchen Zhu, Sean Culatana, Raghuraman Krishnamoorthi, Fanyi Xiao, Yong Jae Lee

*Transactions on Machine Learning Research (TMLR)*, 2025.

**Denoising and Selecting Pseudo-Heatmaps for Semi-Supervised Human Pose Estimation**

**Zhuoran Yu\***, Manchen Wang\*, Yanbei Chen, Paolo Favaro, Davide Modolo

*IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Waikoloa, Hawaii, 2024.

**InPL: Pseudo-labeling the Inliers First for Imbalanced Semi-Supervised Learning**

**Zhuoran Yu**, Yin Li, Yong Jae Lee

*International Conference on Learning Representations (ICLR)*, Kigali, Rwanda, 2023.

**Group R-CNN for Weakly Semi-supervised Object Detection with Points**

Shilong Zhang\*, **Zhuoran Yu**\*, Liyang Liu\*, Xinjiang Wang, Aojun Zhou, Kai Chen

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, New Orleans, 2022.

**Scale-Equalizing Pyramid Convolution for Object Detection**

Xinjiang Wang\*, Shilong Zhang\*, **Zhuoran Yu**, Litong Feng, Wei Zhang

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Virtual, 2020.